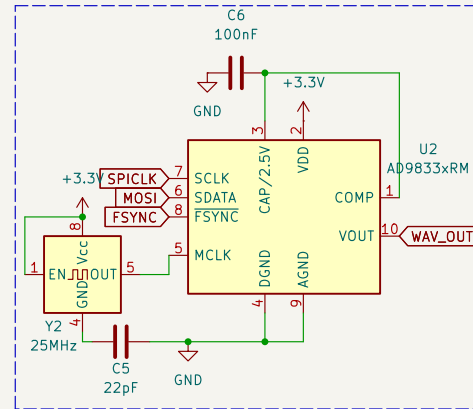
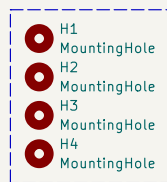
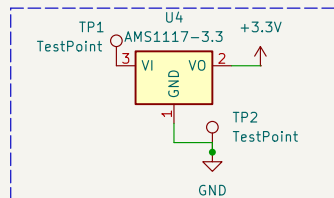


Working:

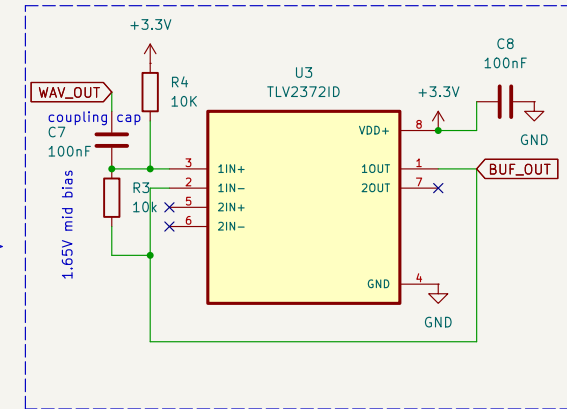
A sine signal generated by the AD9833 is level-shifted and buffered, then applied to a voltage divider formed by a known resistor and an external unknown impedance (Z_{unknown}).

The ADC measures the input voltage (V_{in}) and the divider output (V_{out}), and the unknown impedance is calculated as:

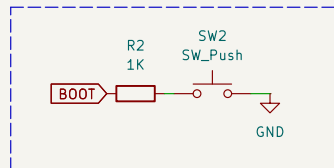
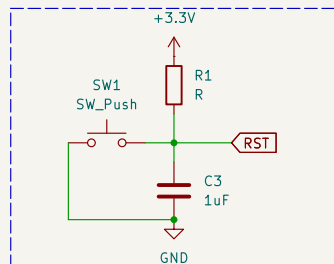
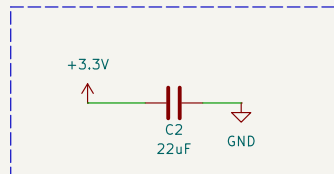
$$Z_{\text{unknown}} = R_{\text{known}} \times (V_{\text{out}} / (V_{\text{in}} - V_{\text{out}}))$$



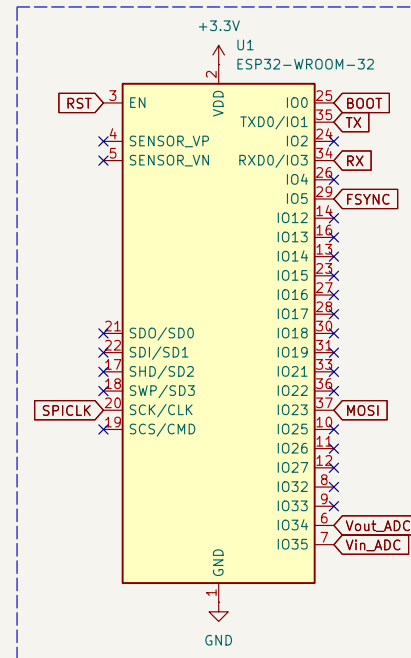
AD9833 Waveform Generator



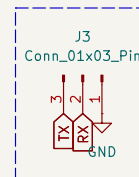
Buffer Stage



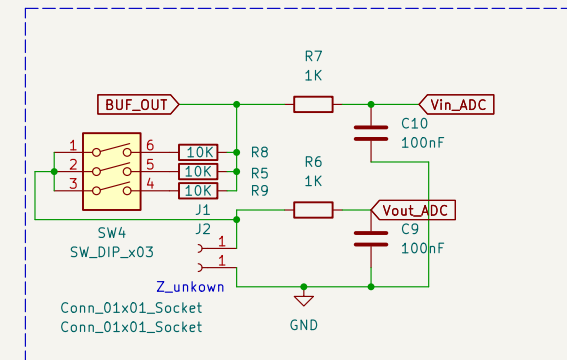
Decoupling, RST, BOOT



ESP32



J3 Conn_01x03_Pin



Voltage Divider + filter before ADC reading

The known resistor has to be switchable to adjust for best resolution for the given Z

Author: Akshat Sharma

Sheet: /
File: potentiostat.kicad_sch

Title:

Size: A4
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Date:

Rev:
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